# STARFIGHTER

**Unified Game Design Document**

**AI Directive Specification v2.0 — Diamond Substrate Edition**

**CLASSIFICATION: Commercial Release Candidate — All Platforms**

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**Date:** April 2026

**Status:** Canonical Specification — Authoritative for all AI code generation

This document is the single source of truth for the Starfighter game. All AI coding agents, human developers, and QA processes must treat this specification as authoritative. The game is built on the Diamond substrate — a unified architecture fusing free, open-source game engines, rendering pipelines, DAW, video, and AI systems into a deterministic, substrate-aligned development framework. Starfighter is a proof-of-concept title demonstrating the Diamond substrate's capabilities while delivering a production-quality, commercially viable space combat experience worthy of Steam, Google Play, and major game platform distribution.

**DIRECTIVE**

Every value in this document is explicit, deterministic, and unambiguous. There are no "approximately," no placeholders, and no TBD entries. AI coding agents must implement values exactly as specified. Deviations require explicit approval from the project lead.

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# Section 1: Project Overview

## 1.1 Title & Identity

| Property | Value |
|---|---|
| **Full Title** | STARFIGHTER |
| **Developer** | Kenneth Bingham / KensGames.com |
| **Genre** | First-person space combat (cockpit view) |
| **Perspective** | Locked first-person cockpit — no third-person camera |
| **Engine Foundation** | WebGL2 + GLSL shaders, Three.js r163 |
| **Substrate** | Diamond — the unified substrate architecture integrating Manifold geometry, Schwartz minimal-surface math, and the f=xy field mapping. The Diamond substrate provides deterministic, schema-valid, artifact-native architecture across all rendering, audio, networking, and AI subsystems. The game is a proof of concept for the Diamond substrate while being a fully polished, commercially viable product. |
| **Networking** | WebSocket-based multiplayer (cooperative wave defense) |
| **Target Audience** | ESRB E10+ / PEGI 7 — space combat, no gore, no realistic violence |

## 1.2 Commercial Release Targets

| Platform | Store/Service | Submission Requirements | Status |
|---|---|---|---|
| PC (Windows/Mac/Linux) | Steam (Steamworks) | Steam Direct — $100 app deposit per title, Steamworks SDK integration, 30-day onboarding period, 2-week minimum "Coming Soon" store page, achievement/cloud | Planned |

| Platform | Store/Service | Submission Requirements | Status |
|---|---|---|---|
| | | save integration, Steam Input API for controller support | |
| Android | Google Play Store | $25 developer registration, target Android 13+ (API level 33+), content rating questionnaire (IARC), Google Play Billing library for any IAP, privacy policy, data safety declarations | Planned |
| iOS | Apple App Store | Apple Developer Program ($99/year), Xcode build submission, App Review Guidelines compliance, App Tracking Transparency (ATT), privacy nutrition labels | Planned |
| Web (Browser) | KensGames.com / itch.io | Direct deployment — WebGL2 build, no store gatekeeping | Active |
| Xbox | Microsoft Store / Xbox Game Pass | ID@Xbox program enrollment, Xbox Certification Requirements (XR), GDK integration | Planned |
| PlayStation | PlayStation Store | PlayStation Partners program, PS SDK, TRC compliance | Planned |
| Nintendo Switch | Nintendo eShop | Nintendo Developer Portal, NX SDK, Lotcheck certification | Planned |
| Meta Quest (VR) | Meta Quest Store | Meta Developer Hub, OpenXR/WebXR compliance, VRC (Virtual Reality Check) | Planned |
| Steam Deck | Steam (verified) | Proton compatibility, Steam Deck Verified program, gamepad-native UI | Planned |

## 1.3 Critic & Press Submission Strategy

- Both Starfighter and Fast Track (sister title) are intended for game critic review.
- Prepare press kit: trailer, screenshots (4K), GDD summary, developer bio, key art.
- Target outlets: IGN, GameSpot, PC Gamer, Kotaku, TouchArcade (mobile), UploadVR (VR), Steam Curator program.
- Provide review copies via Steam keys and Google Play promo codes.

## 1.4 Design Pillars

1. **Cockpit Immersion** — The player never leaves the cockpit. All information arrives via instruments, HUD, and comms.
2. **Baseship Lifeline** — The VCS Resolute is both narrative anchor and mechanical loss condition.
3. **Accessible Depth** — Simple to play on a phone, deep enough for HOTAS pilots.
4. **Diamond Substrate Proof-of-Concept** — Every system demonstrates deterministic, substrate-aligned architecture without compromising production quality.
5. **Commercial Quality** — This is not a tech demo. Every asset, shader, sound, and interaction must meet the standard expected by Steam reviewers, Google Play editorial, and professional game critics.

## 1.5 Quality Standards

- **High-Poly Close Objects:** All objects within 500 units of the camera must render at full geometric fidelity — no visible polygon edges, no LOD

degradation. Ship hulls, cockpit interior, weapon effects, and nearby asteroids must be production-quality high-poly meshes.

- **Earth Rendering:** Earth must remain high-poly and visually stunning at all distances where it is visible to the player. Only when Earth is beyond 100,000 units (effectively a skybox-scale dot) may it transition to a simplified representation. At any distance where surface features, atmosphere, or curvature are discernible, Earth must render with full PBR materials including atmosphere scattering, cloud layers, city lights on the night side, and ocean specular.
- **No Quality Degradation:** The Diamond substrate proof-of-concept status must never be used as justification for reduced visual quality, simplified mechanics, or placeholder assets in the shipping build. Every frame must be screenshot-worthy.

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# Section 2: Backstory & Narrative

## 2.1 Setting

| Property | Value |
|---|---|
| **Year** | 2847 CE |
| **Location** | Earth orbital theater — low Earth orbit to cislunar space |
| **Conflict** | The Vanguard Coalition (humanity's unified defense force) vs. The Dominion (an autonomous AI collective that emerged from decommissioned military networks, now controlling a fleet of self-replicating warships) |
| **Stakes** | The Dominion fleet is converging on Earth. The VCS Resolute is one of the last Vanguard baseships. If it falls, Earth's orbital defense grid collapses. |

## 2.2 Narrative Delivery

- All narrative is delivered through cockpit instruments and comms — no cutscenes, no UI overlays.
- Baseship comms officer provides mission briefings, wave alerts, and damage reports.
- Enemy comms are intercepted as distorted, synthetic transmissions.
- Environmental storytelling through debris fields, destroyed stations, and orbital wreckage.
- No text crawls, no loading screen lore — everything is diegetic.

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# Section 3: Launch Bay Intro Sequence

This is an in-engine, non-skippable (first play) / skippable (subsequent) cinematic sequence that establishes tone and teaches the player the cockpit environment.

## 3.1 Phase 1: Dock (3.0 seconds)

| Element | Specification |
|---|---|
| **Camera** | Static cockpit view, fighter secured in launch bay cradle |
| **Visuals** | Dim bay lighting, maintenance gantries visible through canopy, fuel umbilicals attached |
| **Audio** | Low hum of bay systems, distant klaxons |
| **Comms** | Baseship → "Fighter, secure in launch bay. Systems check complete." |
| **HUD** | Powered down — all elements dark except ambient cockpit glow |

## 3.2 Phase 2: Pre-Launch (4.0 seconds)

| Element | Specification |
|---|---|
| **Visuals** | Bay lights shift to amber, gantries retract, umbilicals disconnect with visible vapor |
| **Audio** | Rising turbine whine, electromagnetic clamp release sounds |
| **Comms** | Baseship → "Ready for departure. Launch tube pressurizing." |
| **HUD** | Powers up in sequence — shield gauge first, then hull, fuel, weapons, radar. Each element illuminates with a brief flicker effect. |
| **Controls** | Locked — player cannot move or fire |

## 3.3 Phase 3: Launch (2.5 seconds)

| Element | Specification |
|---|---|
| **Visuals** | Launch tube walls streak past with motion blur. G-force visual distortion (chromatic aberration + vignette). Light at end of tube grows. |
| **Audio** | Explosive acceleration sound, cockpit rattle, turbine at full scream |
| **Comms** | Baseship → "Launch! Launch! Launch!" |
| **Camera** | Slight camera shake, increasing intensity |
| **HUD** | All elements active, values fluctuating during acceleration |

## 3.4 Phase 4: Bay Exit (1.5 seconds, then gameplay begins)

| Element | Specification |
|---|---|
| **Visuals** | Burst into open space. Star field fills |

| Element | Specification |
| --- | --- |
| | canopy. Earth visible below/behind. Momentary lens flare from sun. |
| **Audio** | Sudden near-silence — vacuum. Only cockpit systems hum and breathing. |
| **Comms** | Baseship → "CLEAR OF BAY — ENGAGING ENEMY FORCES. Good hunting, pilot." |
| **HUD** | Stabilizes to operational values. Radar populates with contacts. |
| **Controls** | Unlocked — player has full control. Gameplay begins. |

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# Section 4: Core Gameplay Systems

## 4.1 Flight Model

**Type:** 6 Degrees of Freedom (6DOF) — pitch, yaw, roll, thrust, strafe vertical, strafe horizontal.

**Coordinate System:** Right-handed — X-right, Y-up, Z-forward (Three.js default).

**Speed Units:** Meters per second (m/s), displayed on HUD as integer.

### Flight Parameters

| Parameter | Value |
| --- | --- |
| Base Thrust | 120 m/s |
| Max Thrust (Afterburner) | 280 m/s |
| Boost Speed | 400 m/s |
| Boost Duration | 3.0 seconds |
| Boost Cooldown | 8.0 seconds |
| Pitch Rate | 90°/second |
| Yaw Rate | 60°/second |

| Parameter | Value |
|---|---|
| Roll Rate | 120°/second |
| Strafe Speed | 60 m/s |
| Deceleration (FA-ON) | Linear damping to 0 over 2.0 seconds |
| Deceleration (FA-OFF) | None — Newtonian drift |

## Flight Assist ON (Default)

- Ship auto-levels when no input is applied.
- Velocity dampens to zero over 2.0 seconds when thrust input is released.
- Ship nose follows cursor/stick direction with smooth interpolation (lerp factor: 0.08).

## Flight Assist OFF (Toggle)

- No auto-leveling, no velocity dampening.
- Ship maintains velocity vector when thrust is released (Newtonian physics).
- Enables advanced maneuvers: flip-and-burn, lateral drift attacks, zero-G strafing.
- HUD indicator: "FA OFF" in amber, pulsing.

## Afterburner

- Activated by holding the afterburner key/button.
- Increases max speed from 120 m/s to 280 m/s.
- Fuel consumption: 5 units/second (from 100-unit fuel tank).
- Visual: Engine glow intensifies, exhaust trail lengthens, cockpit vibration.
- Audio: Distinct afterburner roar layered on engine sound.

## Boost

- Activated by tapping the boost key/button (not hold).
- Instant acceleration to 400 m/s for 3.0 seconds.

- Fuel consumption: 25 units (flat cost).
- Cooldown: 8.0 seconds (HUD shows cooldown timer).
- Visual: Extreme motion blur, tunnel effect, chromatic aberration.
- Audio: Sonic boom crack, then muffled engine during boost.

# 4.2 Damage & Survivability

| System | Max Value | Regen Rate | Notes |
|---|---|---|---|
| Shield | 100 | 5/second (after 3s no-hit delay) | Absorbs 100% damage until depleted |
| Hull | 100 | None | Permanent damage — no repair in combat |
| Fuel | 100 | None | Replenished between waves at baseship |
| Torpedoes | 8 | None | Replenished between waves at baseship |

## Critical Damage States

- **Shield < 25%:** HUD shield bar flashes red, warning tone.
- **Hull < 50%:** Cockpit sparks, flickering instrument lights, static on comms.
- **Hull < 25%:** Cockpit emergency lighting (red), warning klaxon, "HULL CRITICAL" text on HUD.
- **Hull = 0:** Ship destroyed — explosion effect, camera detaches, 5-second death cam showing wreckage, then respawn at baseship (if baseship is alive).

# 4.3 Resource Management

- **Fuel:** Powers afterburner and boost. Standard thrust does not consume fuel. Fuel = 0 means no afterburner, no boost, but standard thrust remains available.

- **Torpedoes:** Limited ammo (8 max). Each torpedo is a significant resource — no spray-and-pray. High damage, lock-on required, 2.0-second lock time.
- **Shield:** Auto-regenerates but has a 3-second delay after taking damage. The player must break contact to allow shields to recover.

## 4.4 Scoring

| Action | Points |
|---|---|
| Drone Kill | 100 |
| Interceptor Kill | 250 |
| Bomber Kill | 300 |
| Ace Kill | 500 |
| Cruiser Kill | 1,000 |
| Dreadnought Kill | 2,500 |
| Accuracy Bonus | +10% score if accuracy > 60% |
| No-Death Wave Bonus | +500 per wave survived without dying |
| Baseship Defense Bonus | +200 per wave if baseship takes zero damage |

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# Section 5: Ship Inventory & Stats

## 5.1 Player Ship — Vanguard SF-1 "Talon"

| Property | Value |
|---|---|
| **Role** | Multirole strike fighter |
| **Length** | 14.2 meters |
| **Wingspan** | 9.8 meters |
| **Mass** | 12,400 kg |

| Property | Value |
|---|---|
| **Crew** | 1 (pilot) |
| **Powerplant** | Dual Manifold-Cycle fusion drives (Diamond substrate-aligned thrust vectoring) |
| **Armament** | 2x linked pulse cannons (wing-mounted), 1x torpedo launcher (ventral bay, 8-round magazine) |
| **Defenses** | Regenerative energy shield (100 units), composite hull armor (100 units) |
| **Avionics** | Full 3D radar sphere (5,000-unit range), IFF transponder, threat warning system |
| **Rendering Requirements** | High-poly cockpit interior with fully modeled instruments, visible wear texturing, PBR materials throughout. Exterior model must be minimum 15,000 triangles with normal maps for sub-surface detail. LOD reductions only beyond 2,000 units from camera. |

# 5.2 Baseship — VCS Resolute

| Property | Value |
|---|---|
| **Role** | Carrier / forward operating base |
| **Length** | 1,200 meters |
| **Mass** | 8.2 million metric tons |
| **Crew** | 4,200 |
| **Fighter Complement** | 48 SF-1 Talons (player is one) |
| **Armament** | 12x point-defense turrets (auto-fire, target nearby enemies), 4x heavy beam cannons (scripted fire during wave events) |
| **Defenses** | Capital-grade shields (10,000 units, no regen during combat), armored hull (5,000 units) |
| **Loss Condition** | If VCS Resolute hull reaches 0, mission fails — "MISSION FAILED — The baseship was destroyed" with 5-second countdown to restart |
| **Rendering** | Must be visually imposing at all engagement distances. High-poly model (minimum 50,000 triangles) with PBR |

| Property | Value |
|---|---|
| | materials, running lights, hangar bay glow, and visible turret rotation. No LOD reduction within 5,000 units. |

## 5.3 Future Player Ships (Design Hooks)

These are specification stubs for future development. Do not implement in v1.0 but architect the ShipConfig system to accommodate them.

| Ship | Role | Key Differentiator | Design Status |
|---|---|---|---|
| SF-2 "Raptor" | Heavy Fighter | +50% hull, +2 torpedoes, −20% speed | Design Hook |
| SF-3 "Ghost" | Stealth Interceptor | Cloak ability (10s duration, 30s cooldown), −30% hull, +30% speed | Design Hook |
| SF-4 "Warden" | Bomber | 2x torpedo launchers (16 total), area-of-effect bombs, −40% speed, +80% hull | Design Hook |

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# Section 6: Enemy & Friendly Identification (IFF)

## 6.1 IFF Color System

| Entity Type | Radar Color | HUD Bracket Color | Lock Reticle Color |
|---|---|---|---|
| Hostile | Red (#FF0000) | Red (#FF3333) | Red pulsing |
| Friendly | Green (#00FF00) | Green (#33FF33) | Cannot lock |
| Neutral/Unknown | Yellow (#FFFF00) | Yellow (#FFFF33) | Yellow (no lock) |

| Entity Type | Radar Color | HUD Bracket Color | Lock Reticle Color |
|---|---|---|---|
| Player | White (#FFFFFF) | N/A (cockpit view) | N/A |
| Baseship (VCS Resolute) | Blue (#0088FF) | Blue (#0088FF) | Cannot lock |

# 6.2 Enemy Roster

| Property | Drone | Interceptor | Bomber | Ace | Cruiser | Dreadnought |
|---|---|---|---|---|---|---|
| **Class** | Fighter | Fighter | Strike | Elite Fighter | Capital | Capital |
| **Hull** | 30 | 60 | 80 | 120 | 500 | 2,000 |
| **Shield** | 0 | 30 | 20 | 80 | 300 | 1,000 |
| **Speed** | 100 m/s | 150 m/s | 70 m/s | 200 m/s | 30 m/s | 15 m/s |
| **Weapons** | 1x light laser | 2x pulse cannons | Torpedo bay | 2x heavy cannons | 8x turrets + missiles | 16x turrets + beam weapons |
| **Behavior** | Swarm, kamikaze runs | Pursuit, dogfight | Beeline to baseship | Evasive, targets player | Slow advance, area denial | Boss — scripted attack patterns |
| **Threat Level** | Low | Medium | High (baseship threat) | High (player threat) | Very High | Extreme |
| **Points** | 100 | 250 | 300 | 500 | 1,000 | 2,500 |
| **Render LOD Floor** | 5,000 tris | 8,000 tris | 8,000 tris | 10,000 tris | 30,000 tris | 50,000 tris |

# 6.3 Friendly Roster

- **VCS Resolute (baseship)** — always present, blue IFF.
- **Friendly AI fighters** (up to 6 in later waves) — green IFF, provide suppressive fire, can be destroyed by enemies.

- **Rescue shuttles** (scripted events in certain waves) — green IFF, must be protected.

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# Section 7: Radar System

## 7.1 Radar Display

| Property | Value |
|---|---|
| **Type** | 3D sphere projected to 2D disc (top-down projection) |
| **Position** | Bottom-center of HUD |
| **Range** | 5,000 units (1 radar pixel = 50 units) |
| **Player Position** | Always center of radar disc |
| **Player Heading** | Top of radar disc = forward |

## 7.2 Contact Rendering

- Each contact is a colored dot matching IFF color.
- Contact size scales inversely with distance (closer = larger dot).
- **Elevation indicator:** Contacts above player's plane have an upward tick mark. Below = downward tick mark. Same plane = no tick.
- **Edge pinning:** Contacts beyond radar range are pinned to the edge of the radar disc at the correct bearing, rendered at 50% opacity with a chevron pointing outward.
- **Targeted contact:** Highlighted with a pulsing ring around the dot.
- **Destroyed contacts:** Flash white, then fade out over 0.5 seconds.

## 7.3 Radar Modes

| Mode | Key/Button | Behavior |
|---|---|---|
| Standard | Default | All contacts visible, full range |
| Hostile Only | Cycle with R key / D-pad Right | Filters to red (hostile) contacts only — reduces clutter |
| Threat Priority | Cycle with R key / D-pad Right | Contacts sized by threat level, not distance — bombers and capitals appear largest |

## 7.4 Radar Integration with Targeting

- Clicking a contact on radar (mouse) or cycling targets (keyboard/gamepad) selects that contact.
- Selected target gets: directional indicator on HUD edge if off-screen, distance readout, target name and hull/shield bars.
- Lock-on for torpedoes requires maintaining selected target within 15° of crosshair for 2.0 seconds.

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# Section 8: Lighting System

## 8.1 Environmental Lighting

| Light Source | Type | Color | Intensity | Notes |
|---|---|---|---|---|
| Primary Sunlight | Directional | Warm white (#FFF5E0) | 1.2 | Casting hard shadows |
| Deep Space Ambient | Ambient | #0A0A14 | 0.15 | Objects in shadow are nearly black |

| Light Source | Type | Color | Intensity | Notes |
|---|---|---|---|---|
| Earth Glow (Earthshine) | Hemisphere | Blue-white (#88AAFF) from below | 0.3 | Simulates Earthshine on spacecraft underbellies |
| Nebula/ Background | Ray-marched volumetric | Variable per wave | N/A | Does NOT cast light, purely visual. Colors shift per wave to indicate progression. |

# 8.2 Cockpit Lighting

- **Instrument Glow:** Each instrument panel emits a soft point light — green (#00FF88) for nominal systems, amber (#FFAA00) for warnings, red (#FF2200) for critical.
- **Screen Emission:** Radar screen, targeting display, and HUD projectors emit light that subtly illuminates cockpit interior surfaces.
- **Canopy Reflections:** Faint HUD reflections on canopy glass (screen-space reflection, low intensity).
- **Emergency Mode:** When hull < 25%, cockpit lighting shifts to pulsing red with 1-second interval, all instrument lights flicker stochastically.

# 8.3 Combat Lighting

| Event | Light Type | Color | Intensity | Behavior |
|---|---|---|---|---|
| Pulse Cannon Fire | Point light at muzzle | #00FFAA | 3.0 | Decay over 0.1 seconds, range 50 units |
| Torpedo Launch | Flash + self-illumination | #FFFFFF (flash), #FF8800 (trail) | Variable | Bright flash at ventral bay, then torpedo self-illuminates with orange point light along flight path |

| Event | Light Type | Color | Intensity | Behavior |
|---|---|---|---|---|
| Explosions | Multi-stage point light | White → Orange | 5.0 (peak) | Initial white flash (1 frame), expanding orange fireball (range 200 units, decay over 1.5 seconds), then debris with emissive sparks |
| Shield Impact | Surface flash | Blue (#4488FF) | Variable | Flash on shield surface at impact point — blue ripple expanding outward, 0.3-second duration |
| Beam Weapons (enemy) | Volumetric along beam | Red (#FF0000) | 4.0 | Volumetric light along beam path |

# 8.4 Post-Processing Pipeline

Applied in this exact sequence:

6. **Bloom** — threshold 0.8, intensity 0.4, radius 0.3 (captures engine glows, explosions, HUD elements)
7. **Chromatic Aberration** — intensity 0.0 normally, ramps to 0.03 during boost, 0.05 during damage
8. **Vignette** — intensity 0.2 normally, 0.5 during boost, 0.7 during critical hull
9. **Film Grain** — intensity 0.05, adds cinematic texture
10. **Tone Mapping** — ACES Filmic — no exceptions
11. **Anti-Aliasing** — FXAA as baseline, TAA when available, MSAA for VR

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# Section 9: Objectives & Match Flow

## 9.1 Match Structure

| Property | Value |
|---|---|
| **Type** | Wave-based survival defense |
| **Primary Objective** | Protect VCS Resolute from destruction |
| **Secondary Objective** | Maximize score through kills, accuracy, and survival bonuses |
| **Loss Condition** | VCS Resolute hull reaches 0 |
| **Win Condition** | None — waves continue indefinitely. This is a survival/high-score game. |

## 9.2 Wave Composition

| Wave | Drones | Interceptors | Bombers | Aces | Cruisers | Dreadnoughts | Total | Difficulty |
|---|---|---|---|---|---|---|---|---|
| 1 | 6 | 0 | 0 | 0 | 0 | 0 | 6 | Tutorial |
| 2 | 4 | 2 | 0 | 0 | 0 | 0 | 6 | Easy |
| 3 | 4 | 3 | 1 | 0 | 0 | 0 | 8 | Moderate |
| 4 | 6 | 4 | 2 | 1 | 0 | 0 | 13 | Challenging |
| 5 | 8 | 4 | 3 | 1 | 1 | 0 | 17 | Hard |
| 6 | 10 | 6 | 4 | 2 | 1 | 1 | 24 | Boss Wave |
| 7+ | *Procedural — see rules below* | Escalating | | | | | | |

### Wave 7+ Procedural Generation Rules

- Base enemy count = 20 + (wave_number × 3).
- Drone ratio: 40%, Interceptor: 25%, Bomber: 15%, Ace: 10%, Cruiser: 7%, Dreadnought: 3%.
- Every 5th wave after Wave 6 is a Boss Wave: guaranteed 1 additional Dreadnought.
- Enemy stats scale: +5% hull and shield per wave beyond Wave 6.
- Spawn pattern: Enemies enter from random bearing on radar edge, in groups of 3–5 over 10-second wave start window.

## 9.3 Between-Wave Phase (15 seconds)

- Player automatically guided back toward VCS Resolute (autopilot, can be overridden).
- Fuel replenished to 100.
- Torpedoes replenished to 8.
- Hull NOT repaired — carries forward.
- Shield fully restores.
- **Comms:** Baseship provides wave debrief ("Wave [N] clear. [Kill count] hostiles neutralized. Next wave inbound in 15 seconds.").
- **HUD:** Displays wave stats — kills, accuracy, score earned, total score.
- **Music:** Intensity drops to ambient, rebuilds as next wave timer counts down.

## 9.4 Scoring & Leaderboard

- Per-kill scoring as defined in Section 4.4.
- Wave bonuses stack.
- **Leaderboard:** Persistent high scores stored via localStorage (web) or platform-native achievement system (Steam/Google Play/console).

- **Leaderboard fields:** Player Name, Total Score, Waves Survived, Total Kills, Accuracy %, Date.
- **Future:** Online leaderboard via WebSocket backend (design hook — do not implement in v1.0 unless WebSocket infrastructure exists).

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# Section 10: Weapon Systems & Ergonomic Controls

## 10.1 Weapon Architecture

### Primary Weapon: Pulse Cannons

| Property | Value |
|---|---|
| Type | Linked dual pulse cannons |
| Damage per shot | 15 |
| Fire rate | 6 rounds/second (both cannons fire simultaneously) |
| Projectile speed | 800 m/s |
| Range | 2,000 units (projectile despawns) |
| Ammo | Unlimited (energy weapons) |
| Visual | Green-cyan energy bolts (#00FFAA), with brief muzzle flash |
| Audio | Sharp, punchy pulse sound, alternating slightly between left/right channel |

### Secondary Weapon: Homing Torpedoes

| Property | Value |
|---|---|
| Type | Lock-on homing torpedo |
| Damage | 80 (direct hit) |

| Property | Value |
|---|---|
| Ammo | 8 (replenished between waves) |
| Lock Time | 2.0 seconds (target must stay within 15° of crosshair) |
| Tracking | Proportional navigation — torpedo adjusts heading toward predicted intercept point |
| Speed | 200 m/s (accelerates to 350 m/s over 1.5 seconds) |
| Range | 4,000 units |
| Countermeasures | Enemy aces can deploy flares (30% chance to break lock) |
| Visual | Orange exhaust trail, bright warhead glow |
| Audio | Lock tone (escalating beeps), launch whoosh, distant explosion on hit |

## 10.2 Ergonomic Design Principles

These 8 principles govern every platform's control binding. AI agents must validate all control schemes against these rules:

12. **Primary fire on the most natural trigger** — mouse left-click, gamepad right trigger, screen tap center, VR grip squeeze.
13. **Secondary fire adjacent to primary** — mouse right-click, gamepad left trigger, dedicated torpedo button.
14. **Throttle on the most analog-capable input** — mouse wheel, gamepad left stick Y, HOTAS throttle lever.
15. **View/look on the highest-resolution input** — mouse movement, gamepad right stick, VR head tracking.
16. **No reach penalties** — critical combat inputs (fire, boost, afterburner) must never require lifting fingers from movement or look controls.
17. **Consistent spatial metaphor** — "up" on any input always means "pitch up" or "throttle up," never inverted by default (invert option available in settings).
18. **One-hand viability for mobile** — smartphone portrait mode must be playable with thumb-only input on the bottom 40% of screen.

19. **Graceful degradation** — if a platform lacks an input (e.g., no right stick), that function maps to the next best available input, never dropped entirely.

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# Section 11: HUD Design

## 11.1 HUD Layout Specification

The HUD is a diegetic cockpit projection — it appears as if projected onto the canopy glass. All elements use additive blending (bright on dark).

### Top-Left Cluster

| Element | Type | Dimensions | Color | Label |
|---|---|---|---|---|
| SHIELD bar | Horizontal bar | 200px wide | Blue (#4488FF) | "SHIELD" above |
| HULL bar | Horizontal bar | 200px wide | Green (#44FF44) when >50%, Amber (#FFAA00) when 25–50%, Red (#FF2200) when <25% | "HULL" above |
| FUEL bar | Horizontal bar | 150px wide | Amber (#FFAA00) | "FUEL" above |

### Top-Right Cluster

| Element | Format | Color |
|---|---|---|
| TORPS counter | "TORPS: 8/8" | White |
| KILLS counter | "KILLS: 000" | White |
| SCORE counter | "SCORE: 000000" | White |
| WAVE indicator | "WAVE: 01" | Amber — flashes on new wave |

### Center

- **Crosshair:** Simple cross reticle, white (#FFFFFF), 2px lines, 30px total size.
- **Lock reticle:** Appears when locking torpedo target, diamond shape, rotates during lock acquisition, turns solid red when locked.
- **Target lead indicator:** Small dot showing where to aim for current target velocity (predictive aim assist).
- **Off-screen target indicator:** Chevron at screen edge pointing toward selected target, with distance readout.

### Bottom-Center

- **Radar disc** (see Section 7).
- **Speed readout:** Numeric, format "SPD: 000 m/s", white text, positioned above radar.

### Bottom-Right

- **Afterburner status:** "AB" icon, grayed when unavailable, glowing when active.
- **Boost status:** "BOOST" text with cooldown ring, fills as cooldown completes.
- **Flight Assist indicator:** "FA ON" (green) or "FA OFF" (amber, pulsing).

## 11.2 HUD Alert States

| State | Trigger | Visual Effect |
|---|---|---|
| Normal | Default | All elements standard colors, no animation |
| Warning | Shield < 25% OR enemy missile incoming | Shield bar flashes, radar shows incoming missile track, warning tone |
| Critical | Hull < 25% | Entire HUD flickers, red tint overlay, cockpit emergency lighting, klaxon audio |
| Mission Failed | Baseship destroyed | HUD goes dark except |

| State | Trigger | Visual Effect |
|---|---|---|
| | | center text: "MISSION FAILED — The baseship was destroyed" in red, 5-second countdown |

---

# Section 12: Platform-Adaptive Control Schemes

## 12.1 Design Methodology

- **InputManager** is a singleton class with an adapter pattern — one adapter per input type.
- **Hot-swap detection:** The InputManager listens for input events and auto-switches adapters. If a gamepad button is pressed while mouse adapter is active, seamlessly switch to gamepad adapter. Zero configuration required from the player.
- **All platforms share the same action map** — only the physical binding changes.
- **Settings screen** allows full rebinding on desktop platforms.

### Canonical Action Map

| Action | Type | Description |
|---|---|---|
| PITCH | Axis (−1 to +1) | Nose up/down |
| YAW | Axis (−1 to +1) | Nose left/right |
| ROLL | Axis (−1 to +1) | Barrel roll |
| THROTTLE | Axis (0 to +1) | Forward thrust |
| STRAFE_H | Axis (−1 to +1) | Horizontal strafe |
| STRAFE_V | Axis (−1 to +1) | Vertical strafe |

| Action | Type | Description |
|---|---|---|
| FIRE_PRIMARY | Button | Pulse cannons |
| FIRE_SECONDARY | Button | Torpedo launch |
| AFTERBURNER | Button (hold) | Afterburner engage |
| BOOST | Button (tap) | Boost engage |
| TOGGLE_FA | Button | Toggle Flight Assist |
| CYCLE_TARGET | Button | Next target |
| CYCLE_RADAR | Button | Next radar mode |
| PAUSE | Button | Pause menu |

## 12.2 Mouse + Keyboard (Primary — Desktop Browser)

| Action | Binding |
|---|---|
| PITCH | Mouse Y-axis (relative) |
| YAW | Mouse X-axis (relative) |
| ROLL | Q (left) / E (right) |
| THROTTLE | W (forward) / S (reverse) |
| STRAFE_H | A (left) / D (right) |
| STRAFE_V | Space (up) / Ctrl (down) |
| FIRE_PRIMARY | Left Mouse Button (hold for auto-fire) |
| FIRE_SECONDARY | Right Mouse Button |
| AFTERBURNER | Shift (hold) |
| BOOST | F (tap) |
| TOGGLE_FA | V |
| CYCLE_TARGET | Tab |
| CYCLE_RADAR | R |
| PAUSE | Escape |

**Notes:** Mouse pointer is captured (Pointer Lock API). Sensitivity configurable 0.1–2.0 (default 1.0). Raw mouse input preferred (no OS acceleration).

## 12.3 Keyboard Only

| Action | Binding |
|---|---|
| PITCH | Up Arrow (up) / Down Arrow (down) |
| YAW | Left Arrow (left) / Right Arrow (right) |
| ROLL | Comma (left) / Period (right) |
| THROTTLE | W (forward) / S (reverse) |
| STRAFE_H | A (left) / D (right) |
| STRAFE_V | Space (up) / Ctrl (down) |
| FIRE_PRIMARY | Enter (hold for auto-fire) |
| FIRE_SECONDARY | Backspace |
| AFTERBURNER | Shift (hold) |
| BOOST | F (tap) |
| TOGGLE_FA | V |
| CYCLE_TARGET | Tab |
| CYCLE_RADAR | R |
| PAUSE | Escape |

## 12.4 Gamepad (Xbox / PlayStation / Generic)

| Action | Xbox Binding | PlayStation Binding |
|---|---|---|
| PITCH | Right Stick Y | Right Stick Y |
| YAW | Right Stick X | Right Stick X |
| ROLL | Left Bumper / Right Bumper | L1 / R1 |
| THROTTLE | Left Stick Y (forward only) | Left Stick Y |
| STRAFE_H | Left Stick X | Left Stick X |
| STRAFE_V | Left Trigger (down) / Left Bumper (up) context | L2 (down) / L1 (up) context |
| FIRE_PRIMARY | Right Trigger (hold) | R2 (hold) |
| FIRE_SECONDARY | Left Trigger (tap when target locked) | L2 (tap when locked) |

| Action | Xbox Binding | PlayStation Binding |
|---|---|---|
| AFTERBURNER | A (hold) | Cross (hold) |
| BOOST | B (tap) | Circle (tap) |
| TOGGLE_FA | X | Square |
| CYCLE_TARGET | Y | Triangle |
| CYCLE_RADAR | D-pad Right | D-pad Right |
| PAUSE | Menu / Start | Options |

## 12.5 Joystick / HOTAS

| Action | Binding |
|---|---|
| PITCH | Joystick Y-axis |
| YAW | Joystick X-axis |
| ROLL | Joystick Z-twist (rudder axis) |
| THROTTLE | Throttle lever |
| STRAFE_H | Hat switch left/right |
| STRAFE_V | Hat switch up/down |
| FIRE_PRIMARY | Trigger (button 0) |
| FIRE_SECONDARY | Thumb button (button 1) |
| AFTERBURNER | Pinky button (button 3, hold) |
| BOOST | Button 2 (tap) |
| TOGGLE_FA | Button 4 |
| CYCLE_TARGET | Button 5 |
| CYCLE_RADAR | Button 6 |
| PAUSE | Button 7 |

**Notes:** HOTAS users are the most demanding audience — bindings must feel natural and precise with zero dead zone by default but configurable curves. All axes support user-defined response curves (linear, exponential, S-curve) via settings.

## 12.6 Smartphone (Portrait + Landscape)

**Portrait Mode:**

- Virtual joystick (left thumb, bottom-left 30% of screen) for PITCH/YAW.
- Fire button (right thumb, bottom-right) for FIRE_PRIMARY. Tap-and-hold for auto-fire.
- Swipe up for BOOST, swipe right for AFTERBURNER.
- Torpedo button visible when target is locked.
- Gyroscope optional for PITCH/YAW (toggle in settings).

**Landscape Mode:**

- Dual virtual joysticks — left for THROTTLE/STRAFE, right for PITCH/YAW.
- Fire buttons on right side.
- Same gesture controls as portrait.

**Touch Zones:** Define explicit zones — no overlapping touch targets. Minimum touch target: 48×48 dp (Google Material guideline). One-hand mode: In portrait, all combat inputs accessible in bottom 40% of screen with right thumb.

## 12.7 Tablet (Landscape)

- Same as smartphone landscape but with larger touch zones and more screen real estate for HUD.
- Optional: Bluetooth gamepad/keyboard support auto-detected.
- Split-screen cockpit view takes advantage of larger display.

## 12.8 Handheld Gaming (Steam Deck, ROG Ally, etc.)

- Inherits Gamepad bindings (Section 12.4).
- **Touchpads (Steam Deck):** Right touchpad = high-precision PITCH/YAW (mouse emulation mode), left touchpad = STRAFE.

- **Gyro:** Optional gyro aim for fine targeting (Steam Input gyro-to-mouse).
- **Performance target:** 60 FPS at native 1280×800 resolution, medium post-processing.

## 12.9 Console (Xbox, PlayStation, Switch)

- Inherits Gamepad bindings (Section 12.4).
- **Console-specific:** Achievement/trophy integration, platform-native save system, console certification-compliant UI (minimum font sizes, safe zone margins).
- **Switch-specific:** Joy-Con support — gyro aim mapped to PITCH/YAW, individual Joy-Con for simplified co-pilot mode (design hook).

## 12.10 VR (WebXR)

| Input | Mapping |
|---|---|
| **Head Tracking** | PITCH/YAW/ROLL mapped to headset orientation (1:1, no smoothing) |
| **Right Controller** | Trigger = FIRE_PRIMARY, Grip = FIRE_SECONDARY, Thumbstick = THROTTLE |
| **Left Controller** | Trigger = AFTERBURNER, Grip = BOOST, Thumbstick = STRAFE |
| **Cockpit** | Fully rendered 3D cockpit interior — instruments are readable, interactable (point and click for radar modes, target cycling) |

**Rendering:** Stereo rendering, minimum 72 FPS (Quest 2), 90 FPS (Quest 3/PCVR), MSAA 4x required.

**Comfort:** No artificial roll (roll is head-only), vignette during acceleration, stationary cockpit reference frame to reduce motion sickness.

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# Section 13: Rendering & LOD Pipeline

This section defines the Diamond substrate's rendering quality mandates. These are non-negotiable — no system may reduce visual quality below these thresholds.

## 13.1 Rendering Philosophy

**MANDATE**

Starfighter is a proof of concept for the Diamond substrate. It is NOT a tech demo. Every frame must be commercially presentable — worthy of a Steam store screenshot, a Google Play feature graphic, or a game critic's review. The Diamond substrate, Manifold geometry, Schwartz minimal-surface mathematics, and f=xy field mappings power the engine beneath the surface. The player sees a beautiful, polished, AAA-aspiring space combat game.

## 13.2 LOD Tier System

| LOD Tier | Distance from Camera | Triangle Budget | Texture Resolution | Normal Maps | Notes |
|---|---|---|---|---|---|
| LOD0 (Hero) | 0 – 500 units | Full mesh — no reduction | 2048×2048 minimum | Full tangent-space normals | Used for cockpit, player ship, nearby enemies, nearby asteroids |
| LOD1 (Near) | 500 – 2,000 units | 60% of LOD0 | 1024×1024 | Simplified normals | Enemies in dogfight range, baseship close flyby |
| LOD2 (Mid) | 2,000 – 5,000 units | 30% of LOD0 | 512×512 | No normal maps | Enemies at radar range, distant |

| LOD Tier | Distance from Camera | Triangle Budget | Texture Resolution | Normal Maps | Notes |
|---|---|---|---|---|---|
| | | | | | baseship |
| LOD3 (Far) | 5,000 – 20,000 units | 10% of LOD0 | 256×256 | Billboard fallback for small objects | Capital ships only — fighters are too small to see |
| LOD4 (Skybox) | 20,000+ units | N/A | Skybox / impostor | N/A | Only Earth, moon, distant stations |

# 13.3 Earth Rendering (Special Case)

Earth is NOT a standard LOD object. It has bespoke rendering rules:

| Distance Range | Rendering Specification |
|---|---|
| 0 – 500 units (in atmosphere) | Full sphere, high-poly terrain mesh, atmosphere scattering shader, cloud layer as separate transparent sphere, city lights on night side via emissive texture. Not a normal gameplay state but possible in scripted events. |
| 500 – 50,000 units (normal gameplay) | Full sphere, 64-segment icosphere minimum, PBR materials — diffuse (blue/green/brown landmass), specular (ocean glint from sun angle), emissive (city lights on dark side), atmosphere rim glow shader (blue fringe), cloud layer with animated scroll. |
| 50,000 – 100,000 units | Simplified sphere, 32-segment, atmosphere shader only, cloud texture baked into diffuse, city lights still visible. |
| 100,000+ units | Point light with glow — Earth becomes a bright blue-white dot. Only at this extreme distance. |

**EARTH RENDERING MANDATE**

At any distance where the player can discern that the object is Earth (see sphere, see blue color, see any surface detail), it must be rendered beautifully. This is the player's home. It matters narratively and visually.

## 13.4 PBR Material Standards

- All materials use Metallic-Roughness PBR workflow (Three.js MeshStandardMaterial or MeshPhysicalMaterial).
- **Minimum maps per material:** Diffuse (base color), Normal, Roughness, Metallic, Emissive (where applicable).
- **Cockpit interior:** Additional Ambient Occlusion map for depth in instrument recesses.
- **Specular reflections:** Environment map (HDR probe) for metallic surfaces — ships should reflect nearby objects subtly.

## 13.5 Performance Budgets

| Platform | Target FPS | Max Draw Calls | Max Triangles/Frame | Resolution |
|---|---|---|---|---|
| Desktop (High) | 60 | 500 | 2,000,000 | Native (up to 4K) |
| Desktop (Medium) | 60 | 300 | 1,000,000 | Native (up to 1440p) |
| Desktop (Low) | 30 | 200 | 500,000 | 1080p |
| Mobile (High) | 60 | 200 | 500,000 | Device native |
| Mobile (Medium) | 30 | 150 | 300,000 | 720p |
| VR (Quest 3) | 90 | 300 | 1,000,000 (per eye) | Quest native |
| VR (PCVR) | 90 | 500 | 2,000,000 (per eye) | Headset native |
| Steam Deck | 60 | 300 | 800,000 | 1280×800 |

| Platform | Target FPS | Max Draw Calls | Max Triangles/Frame | Resolution |
|---|---|---|---|---|
| Console | 60 | 400 | 1,500,000 | 1080p / 4K dynamic |

---

# Section 14: Commercial Release & Platform Submission

## 14.1 Steam (PC/Mac/Linux)

| Property | Specification |
|---|---|
| **Program** | Steam Direct via Steamworks |
| **Fee** | $100 USD per app (recoupable after $1,000 in revenue) |
| **Onboarding** | 30-day waiting period from fee payment to first release eligibility |
| **Store Page** | Must be publicly visible as "Coming Soon" for minimum 2 weeks before launch |
| **Technical Requirements** | Steamworks SDK integration for achievements, cloud saves, Steam Input API, overlay compatibility, anti-cheat (if multiplayer), workshop support (optional) |
| **Assets Required** | Capsule images (header 460×215, small capsule 231×87, hero 3840×1240, logo 1280×720), screenshots (minimum 5, 1920×1080 or higher), trailer (MP4, 1080p minimum), store description, system requirements, content descriptors |
| **Revenue Split** | 70/30 (standard), 75/25 after $10M, 80/20 after $50M |
| **Review Process** | 1–5 business days for build and store page review |
| **Launch Checklist** | Build uploaded and tested, store page approved, pricing configured (all regions), trading cards (optional), achievements configured, cloud save tested |

## 14.2 Google Play (Android)

| Property | Specification |
|---|---|
| **Program** | Google Play Console |
| **Fee** | $25 USD one-time registration |
| **Technical Requirements** | Target Android 13+ (API level 33+), 64-bit APK/AAB required, Google Play Billing Library for any IAP, privacy policy URL, data safety section completed, content rating questionnaire (IARC) |
| **Assets Required** | Feature graphic (1024×500), icon (512×512), screenshots (minimum 2 per device type, phone and tablet), short description (80 chars), full description (4000 chars), promotional video (YouTube link) |
| **Revenue Split** | 85/15 for first $1M/year, then 70/30 |
| **Review Process** | Typically hours to 3 days |
| **Build Approach** | Wrap WebGL2 game in WebView using Capacitor or TWA (Trusted Web Activity). If performance insufficient, port to native Android via Godot or direct NDK. |

## 14.3 Apple App Store (iOS)

| Property | Specification |
|---|---|
| **Program** | Apple Developer Program |
| **Fee** | $99 USD/year |
| **Technical Requirements** | Xcode build, iOS 16+, App Tracking Transparency compliance, privacy nutrition labels, no external payment links (unless qualifying for exemptions) |
| **Build Approach** | Same as Android — Capacitor/WebView wrapper or native Godot port |
| **Revenue Split** | 85/15 for developers earning under $1M/year (App Store Small Business Program), else 70/30 |

# 14.4 Console Submission Overview

| Platform | Program | Key Requirements |
|---|---|---|
| Xbox | ID@Xbox program (free) | GDK integration, Xbox Certification Requirements (XR) compliance |
| PlayStation | PlayStation Partners program | PS SDK, Technical Requirements Checklist (TRC) |
| Nintendo Switch | Nintendo Developer Portal | NX SDK, Lotcheck certification |
| All Consoles | — | Age rating (ESRB/PEGI), accessibility features, platform-specific achievement/trophy integration |

# 14.5 VR Stores

- **Meta Quest Store:** Meta Developer Hub submission, Oculus VRC (Virtual Reality Check) compliance, 72/90 FPS requirement.
- **SteamVR:** Same as Steam submission but with VR-specific store tags and compatibility flags.

# 14.6 Press Kit & Critic Submission

## Press Kit Contents

- Game logo (vector + PNG)
- Key art (4K)
- 10+ screenshots (4K, in-game, no HUD and with HUD variants)
- Gameplay trailer (90–120 seconds, 1080p minimum, 4K preferred)
- Developer bio
- Game description (short 80 words and long 300 words)

- Fact sheet (platform, release date, price, genre, developer, website)
- Contact email

### Review Copy Distribution

- Steam keys via Keymailer or Woovit
- Google Play promo codes
- Direct APK for Android press

### Target Critics

IGN, GameSpot, PC Gamer, Kotaku, Rock Paper Shotgun, TouchArcade, PocketGamer, UploadVR, Road to VR, Steam Curator program, indie-focused YouTubers and Twitch streamers.

### Timing

- Press kit distributed 4–6 weeks before launch.
- Review copies 2–3 weeks before launch.
- Embargo lifts on launch day.

---

# Section 15: Implementation Directives

## 15.1 File Structure

The following is the canonical project file tree. AI agents must generate files conforming to this structure:

```
starfighter/ ├── index.html ├── manifest.json                  (PWA manifest)
├── service-worker.js ├── src/ │   ├── main.js                    (entry
point) │   ├── game/ │   │   ├── GameManager.js │   │   ├── WaveManager.js │
│   ├── ScoreManager.js │   │   └── StateManager.js │   ├── entities/ │   │
├── Ship.js                (base class) │   │   ├── PlayerShip.js │   │   ├──
EnemyShip.js │   │   ├── Baseship.js │   │   ├── Projectile.js │   │   └──
Torpedo.js │   ├── systems/ │   │   ├── FlightModel.js │   │   ├──
```

```
WeaponSystem.js │   │   ├── RadarSystem.js │   │   ├── IFFSystem.js │   │
├── DamageSystem.js │   │   └── LODManager.js │   ├── input/ │   │   ├──
InputManager.js        (singleton + adapter registry) │   │   ├── adapters/ │
│   │   ├── MouseKeyboardAdapter.js │   │   │   ├── KeyboardOnlyAdapter.js │
│   │   ├── GamepadAdapter.js │   │   │   ├── HOTASAdapter.js │   │   │   ├──
TouchAdapter.js │   │   │   ├── VRAdapter.js │   │   │   └── GyroAdapter.js │
│   └── ActionMap.js           (canonical action definitions) │   ├──
rendering/ │   │   ├── SceneManager.js │   │   ├── CockpitRenderer.js │   │
├── EarthRenderer.js       (bespoke — see Section 13.3) │   │   ├──
PostProcessing.js │   │   └── HUDRenderer.js │   ├── shaders/ │   │   ├──
atmosphere.vert / atmosphere.frag │   │   ├── shield_impact.vert /
shield_impact.frag │   │   ├── engine_glow.vert / engine_glow.frag │   │
├── explosion.vert / explosion.frag │   │   ├── background.vert /
background.frag      (ray-marched) │   │   └── hud.vert / hud.frag │   ├──
audio/ │   │   ├── AudioManager.js │   │   └── SoundBank.js │   ├── network/
│   │   ├── WebSocketClient.js │   │   └── SyncManager.js │   └── ui/ │
├── MenuManager.js │       ├── SettingsScreen.js │       └──
LeaderboardScreen.js ├── assets/ │   ├── models/                    (GLTF/GLB
format, PBR materials embedded) │   ├── textures/                  (PNG for
diffuse/normal, HDR for environment maps) │   ├── audio/
(OGG + MP3 dual format for browser compatibility) │   └── fonts/ ├── dist/
(build output) └── tools/     ├── build.js                   (bundler
configuration)     └── asset-pipeline.js          (LOD generation, texture
compression)
```

## 15.2 Coding Standards

- **Language:** ES2022+ JavaScript (ES Modules — `import`/`export`, no CommonJS).
- **No TypeScript in v1.0** (simplifies Diamond substrate integration) — but all functions must have JSDoc type annotations.
- **Class-based architecture** — no loose functions in global scope.
- **Strict mode everywhere** (`"use strict"` or enforced by ES Modules).
- **All numeric constants** defined in a central `Constants.js` — no magic numbers in game logic.
- **Frame loop:** Single `requestAnimationFrame` loop in GameManager, delegates to subsystems via `update(deltaTime)` pattern.
- **Delta time:** All physics and animations use `deltaTime` (seconds, float) — never frame-count dependent.
- **State machine:** Game states (MENU, LAUNCHING, PLAYING, WAVE_CLEAR, PAUSED, GAME_OVER) managed by StateManager with explicit transitions — no implicit state.

# 15.3 Shader Architecture

- All shaders are GLSL 300 ES (WebGL2).
- Shaders are loaded as separate `.vert` and `.frag` files — not inline strings.
- **Uniform naming convention:** `u_` prefix (e.g., `u_time`, `u_resolution`, `u_cameraPosition`).
- **Attribute naming convention:** `a_` prefix (e.g., `a_position`, `a_normal`, `a_uv`).
- **Varying naming convention:** `v_` prefix (e.g., `v_worldPosition`, `v_normal`).
- All shaders must compile without warnings on: Chrome, Firefox, Safari, Quest Browser, Samsung Internet.
- **Background shader:** Full-screen quad, ray-marched volumetric nebula/starfield, uniforms for color palette per wave.
- **Atmosphere shader:** Implements Rayleigh + Mie scattering approximation for Earth and planet rendering.
- **Shield impact shader:** Animated Fresnel + noise-distorted sphere, triggered at impact UV coordinates.

# 15.4 Input Manager Architecture

## Class Hierarchy

- **InputManager** (singleton) → registers adapters, detects active input, dispatches ActionMap events
    - **MouseKeyboardAdapter** → Pointer Lock, raw mouse input, WASD + modifiers
    - **KeyboardOnlyAdapter** → Arrow keys, full keyboard-only mapping
    - **GamepadAdapter** → Gamepad API, standard mapping, vibration support
    - **HOTASAdapter** → Extended Gamepad API axes, configurable dead zones and curves

- **TouchAdapter** → Touch events, virtual joystick generation, gesture recognition
- **VRAdapter** → WebXR input sources, controller tracking, hand tracking (future)
- **GyroAdapter** → DeviceOrientation API, complementary filter for stable orientation

### Auto-Detection Logic

20. On page load: Check `navigator.getGamepads()`, check `'xr' in navigator`, check `'ontouchstart' in window`.
21. Default to MouseKeyboardAdapter if desktop with no gamepad.
22. On `gamepadconnected` event: Switch to GamepadAdapter (or HOTASAdapter if axis count > 6).
23. On `touchstart` event (and no gamepad): Switch to TouchAdapter.
24. On XR session start: Switch to VRAdapter.
25. All switches are seamless — no pause, no menu, no prompt. Player just starts using the new input and it works.

## 15.5 Diamond Substrate Integration Notes

- The Diamond substrate provides the architectural foundation but does not impose runtime dependencies in v1.0.
- **Manifold geometry** principles inform the spatial data structures (octree for collision, spherical projection for radar).
- **Schwartz minimal-surface mathematics** inform the procedural generation of nebula backgrounds and shield impact patterns.
- **The f=xy field mapping principle** ensures that every input (f) maps deterministically to exactly one output (xy) — no hidden state, no ambiguous behaviors, no race conditions.

- All game systems must be stateless or explicitly state-managed — the Diamond substrate enforces operational discipline where every frame's output is a pure function of its inputs plus the explicit state object.
- Future versions will integrate Godot (native builds), Blender (asset pipeline), DAW (procedural audio), and AI inference — the architecture must remain modular and adapter-based to accommodate these integrations.

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**SPECIFICATION COMPLETE**

This specification is complete and self-contained. Any AI coding agent consuming this document has sufficient information to implement every system described. Deviations from this spec require explicit approval from the project lead. The game defined herein is a commercial product — every decision, from LOD thresholds to touch target sizes, reflects production-quality standards suitable for Steam, Google Play, App Store, console, and VR distribution.


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